

MUHAMMAD ASAD

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EDUCATION

Bachelor of Engineering in Computer Software Engineering

Sep 2024 – Sep 2028

Bahria University – Karachi Campus

Intermediate – Computer Science (FSc)

Completed

Malir Cant college

EXPERIENCE

Software Development Intern (Remote)

2024

CodeAlpha – Project-Based Internship

- Delivered two production-quality Java applications (Student Grade Manager, Hotel Reservation System) under real internship specifications; both reviewed on GitHub.
- Applied clean code principles, OOP design patterns, Git version control, and project documentation throughout a 4-week engagement.
- Both submissions adopted as reference implementations on the internship platform following positive architecture review.

Backend Development Trainee (Remote) June 2026 – Present (3 Months)

ByteCorp Technologies – Training Program

- Collaborating with a technical mentor on the end-to-end development of a production-style backend system, from database design through API implementation and security.
- Following a structured PR-based workflow, submitting code for review and iterating on mentor feedback to meet production-grade engineering standards.
- Taking ownership of independent modules, including schema design, authentication, and authorization, while adhering to industry-standard Git and version control practices.
- Delivered Phase 1 of the project on schedule, with subsequent phases planned to expand into advanced software design patterns.

PROJECTS

Food Heaven Website | *HTML, CSS, JavaScript, Bootstrap 5, .NET Framework* | [GitHub](#)

- Built a premium restaurant management system featuring a modern UI, dynamic menu display, and a robust table reservation system with responsive design.
- Implemented client-side interactivity using vanilla JavaScript and Bootstrap 5 grid for a seamless user experience across all devices.

Algo Puzzle Board | *HTML, CSS, JavaScript, Data Structures & Algorithms, .NET framework* | [GitHub](#)

- Engineered an interactive algorithm visualization platform covering sorting, pathfinding, N-Queens, Knight's Tour, Graph Coloring, TSP, Huffman Coding, Dijkstra's, and Kruskal's MST.
- Designed modular dashboard architecture with animated step-by-step algorithm walkthroughs to aid conceptual understanding.

Requirement Genie | *Python, NLP, AI/ML, Streamlit* | [GitHub](#)

- Developed an AI-driven requirements engineering assistant that automates the generation of SRS (Software Requirements Specification) documents using NLP techniques.
- Integrated a conversational interface for systematic requirements gathering, reducing manual documentation effort significantly.

Job Ranking System (SkillMesh) | *Python, DSA, Greedy Algorithms, OOP* | [GitHub](#)

- Built a DSA-based smart candidate ranking and recommendation engine that scores job applicants using skill matching, experience scoring, and greedy budget optimization algorithms.
- Implemented Max Heap data structure for efficient top-candidate retrieval with an Explain Mode for step-by-step algorithm tracing.

Decode & Discover | *JavaScript, HTML, CSS, Cryptography* | [GitHub](#)

- Developed an educational cryptography tool that teaches core security concepts through interactive encryption and decryption visualizations with a narrative-driven story format.
- Designed three distinct story-based learning modules to engage users while reinforcing cipher and encoding principles.

MIPS Productivity Dashboard | *MIPS Assembly, Computer Architecture* | [GitHub](#)

- Built a low-level task management system entirely in MIPS Assembly language featuring task tracking, calendar view, and direct memory status updates.
- Demonstrated deep understanding of computer architecture, register management, and memory-mapped I/O in a resource-constrained environment.

Snake Game | *Python, OOP, Pygame / JavaScript* | [GitHub](#)

- Implemented a classic retro-style Snake Game with smooth controls, real-time score tracking, collision detection logic, and progressive difficulty scaling.

Python Streamlit Dashboard Suite | *Python, Streamlit, Pandas, Matplotlib* | [GitHub](#)

- Developed 3+ interactive web dashboards with real-time data filtering, dynamic charting, and multi-source dataset aggregation using Pandas and Streamlit widgets.
- Self-taught Streamlit beyond curriculum; published deployable dashboard apps demonstrating data wrangling and visualization skills.

Student Grade Tracker & Manager | *Java, OOP, Collections Framework* | [GitHub](#)

- Built a Java console application with CRUD operations, automated grade computation (average, highest, lowest), and formatted report generation.
- Implemented encapsulation, inheritance, and polymorphism for a clean, extensible OOP architecture; delivered as a CodeAlpha internship project.

Hotel Reservation System | *Java, OOP, ArrayList, Input Validation* | [GitHub](#)

- Designed a hotel management application supporting room booking, guest registration, real-time availability checks, and full reservation lifecycle management.
- Modeled domain entities (Room, Guest, Booking) as Java classes with robust input validation and error handling for production-grade reliability.

Python Mini-Projects Collection (10+) | *Python, OOP, File I/O* | [GitHub](#)

- Delivered 10+ Python applications including To-Do Manager, Calculator, Quiz Engine, and Number Guesser, each reinforcing OOP design, file handling, and clean code practices.

FoodHeaven – Restaurant POS System | *Python, Django REST Framework, PostgreSQL, JavaScript, PyWebView* | [GitHub](#)

- Built a native desktop Point of Sale system for restaurants with a dark-themed register UI, real-time cart management, GST calculation, and a Kitchen Monitor Display for live order tracking.
- Developed an Analytics Dashboard using Chart.js for sales trends, along with printable invoice generation and multi-mode checkout backed by a PostgreSQL database across modular Django apps.

TECHNICAL SKILLS

Languages: C#, Java, Python, JavaScript, MIPS Assembly, SQL, HTML, CSS

Frameworks & Libraries: Bootstrap 5, Streamlit, Pandas, Matplotlib, NumPy, Java Swing, C# Windows Forms, Django, Django REST Framework, PyWebView, Chart.js

Databases: MySQL, SQL Server, SQLite, PostgreSQL, Oracle

Developer Tools: Git, GitHub, VS Code, Visual Studio, IntelliJ IDEA, PyCharm, Netlify

Core Concepts: OOP, Data Structures & Algorithms, Design and Analysis of Algorithms, REST APIs, NLP, Computer Architecture, Agile Basics

Software Architecture & Design: Design Patterns, Design Principles (SOLID), MVC, Client-Server, Layered, Peer-to-Peer, Pipe-and-Filter etc

ACHIEVEMENTS & HIGHLIGHTS

- 14+ open-source projects on GitHub across Java, Python, C#, and JavaScript with a consistent commit history demonstrating real-world development discipline.
- Portfolio deployed on Netlify showcasing 10+ production-quality projects spanning web development, AI tools, algorithm visualizations, and systems programming.
- CodeAlpha Internship: completed a remote software development internship, delivering enterprise-style Java applications under real project deadlines.
- Self-directed learning: independently mastered Streamlit, Pandas, NLP basics, and data visualization tools beyond coursework; shipped multiple deployable applications.
- Built an AI-powered requirements engineering tool (Requirement Genie) demonstrating initiative in applying NLP to real software engineering problems.